Supervised Learning Assignment 1

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Introduction

This reporrt analyses and models median house proces across a range of features inlcluding taxes, property and crime rates. The median house value is regressed against these features using several regression models, namely staight linear regression using all features, backwise step regression, ridge regression and lasso regression. The out of sample mean sqaured errrs are assesed across the the results of these models. The inclusion of an interaction variables, chas_rm, is added to the best performing model*

Exploratory Data Analysis

For the sake of data and analysis completeness, EDA is conducted on the entire Boston dataset.

Data Attributes

Table 1: boston data attributes

row_count	column_count
506	13

Table 2: Boston Data Data Summary

feature	min	q25	mean	meadian	q75	max	sd	datatype	na_count
crim	0.00632	0.082045	3.6135236	0.25651	3.677083	88.9762	8.601744	numeric	0
zn	0.00000	0.000000	11.3636364	0.00000	12.500000	100.0000	23.322521	numeric	0
indus	0.46000	5.190000	11.1367787	9.69000	18.100000	27.7400	6.860029	numeric	0
chas	0.00000	0.000000	0.0691700	0.00000	0.000000	1.0000	0.244949	integer	0
nox	0.38500	0.449000	0.5546951	0.53800	0.624000	0.8710	0.100000	numeric	0
rm	3.56100	5.885500	6.2846344	6.20850	6.623500	8.7800	0.700000	numeric	0
age	2.90000	45.025000	68.5749012	77.50000	94.075000	100.0000	28.148890	numeric	0
dis	1.12960	2.100175	3.7950427	3.20745	5.188425	12.1265	2.104757	numeric	0
rad	1.00000	4.000000	9.5494071	5.00000	24.000000	24.0000	8.707468	integer	0
tax	187.00000	279.000000	408.2371542	330.00000	666.000000	711.0000	168.537118	integer	0
ptratio	12.60000	17.400000	18.4555336	19.05000	20.200000	22.0000	2.165641	numeric	0
lstat	1.73000	6.950000	12.6530632	11.36000	16.955000	37.9700	7.140728	numeric	0
medv	5.00000	17.025000	22.5328063	21.20000	25.000000	50.0000	9.197282	numeric	0

The complete data set contains 506 observations across 13 columns. The data is complete, with no null values, and all columns are registered as numeric. **Tax (28404), age (792), and zn (543)** display the greatest variance across the features, whilst **nox (0.01) and rm (0.49)** show the least variance across their respective values.

Median House Value Distributions

The data data is imbalanced, with more observations for the chas class=0.

The median house value for observations with chas=0 is slightly higher at 23.3K, and the observations in this group skew towards higher values, in contrast to the distribution of the chas=1 observations, which is relatively more balanced around the median. Despite the difference in distributions, the two box plots show some overlap, which could indicate that the difference in means of these two groups may not be statistically significant.

Most observations are contained with the rad values= 4,5 and 24

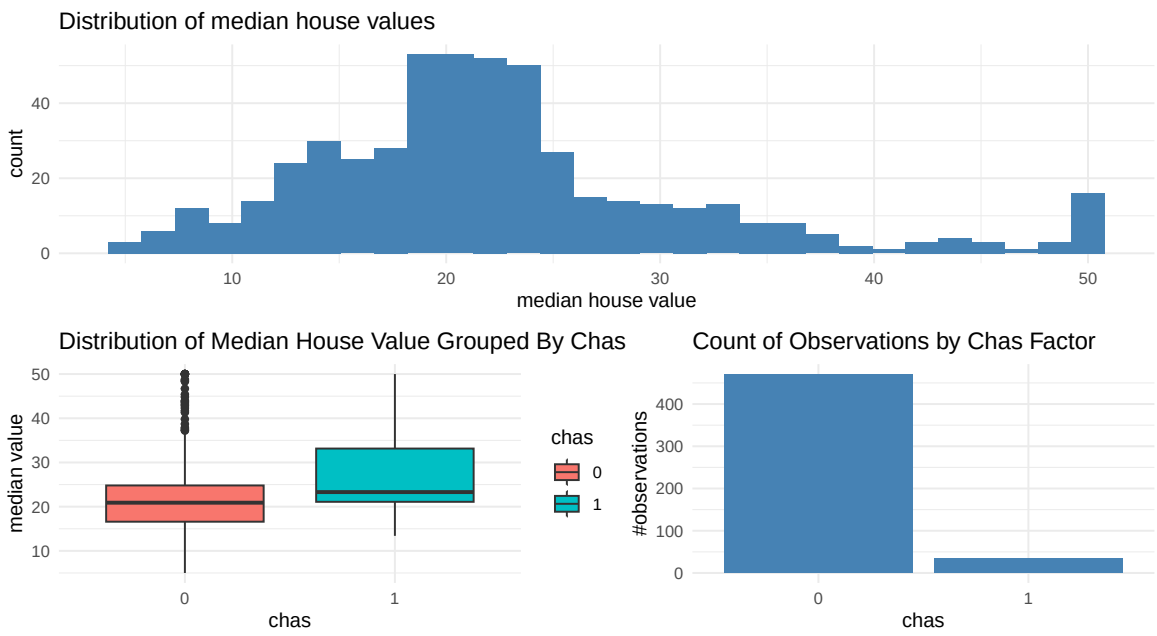


Figure 1: Median House Value Distribution,Grouped by Chas

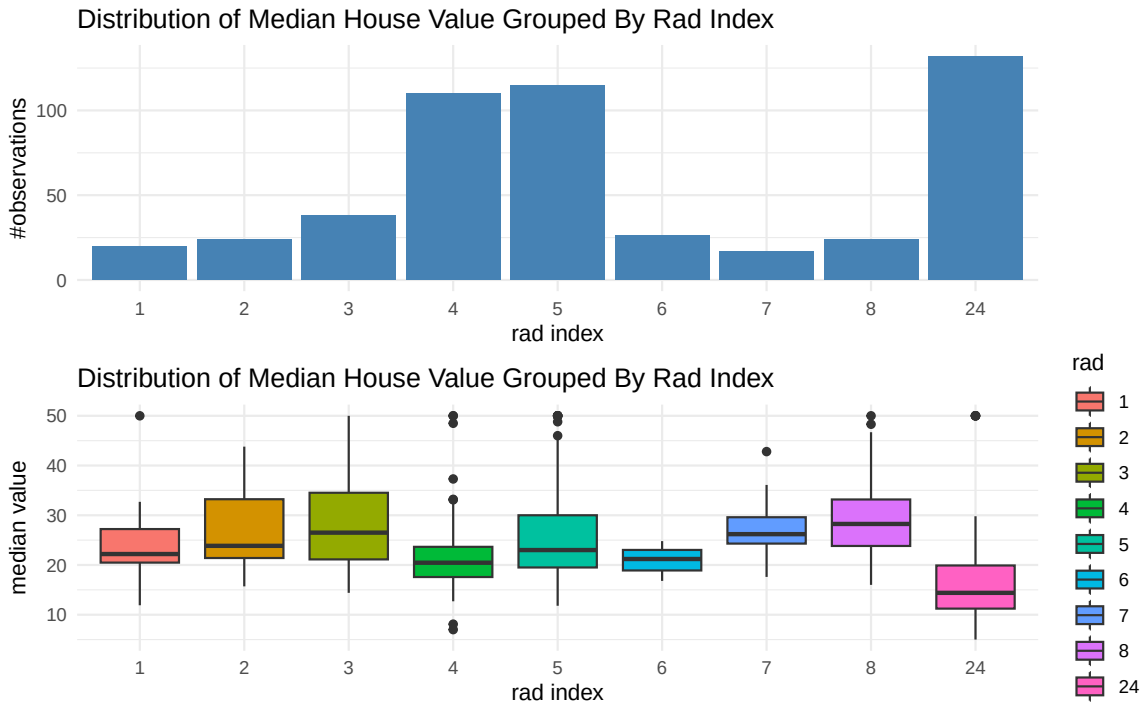


Figure 2: Distribution of Median House Values by Radial Accesibility Index

Grouping house value distributions by the RAD index value shows that median house values do not consistently increase or decrease with changes in the rad index. Observations with a rad value of 24 display the lowest house values among the index groups.

Feature Correlations

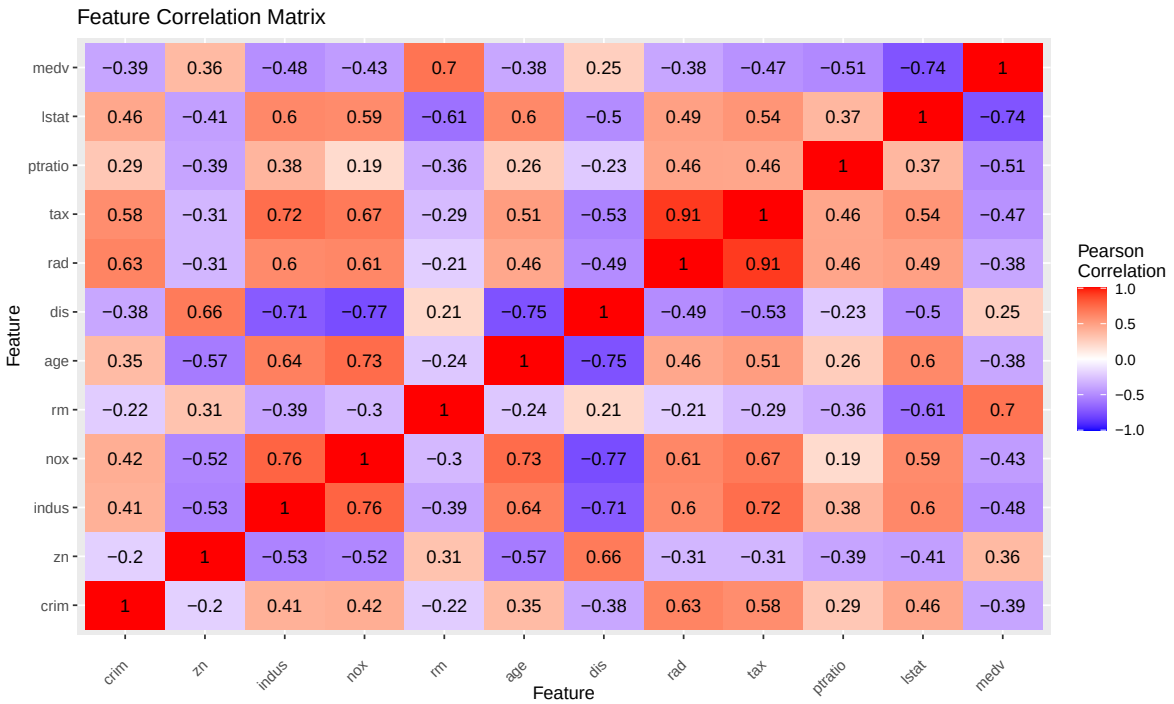


Figure 3: Feature Correlations

The data shows evidence of multicollinearity, with 8 out of 11 explanatory features reporting linear correlations >=70%. In particular, features, **rm and tax** display the largest correlation (~91%), in absolute value. Features, **lstat (~-74%)** and **rm (~70%),** show the highest absolute value correlations to the response value variable. *this could what is causing our variance to blow up-talk about the implications

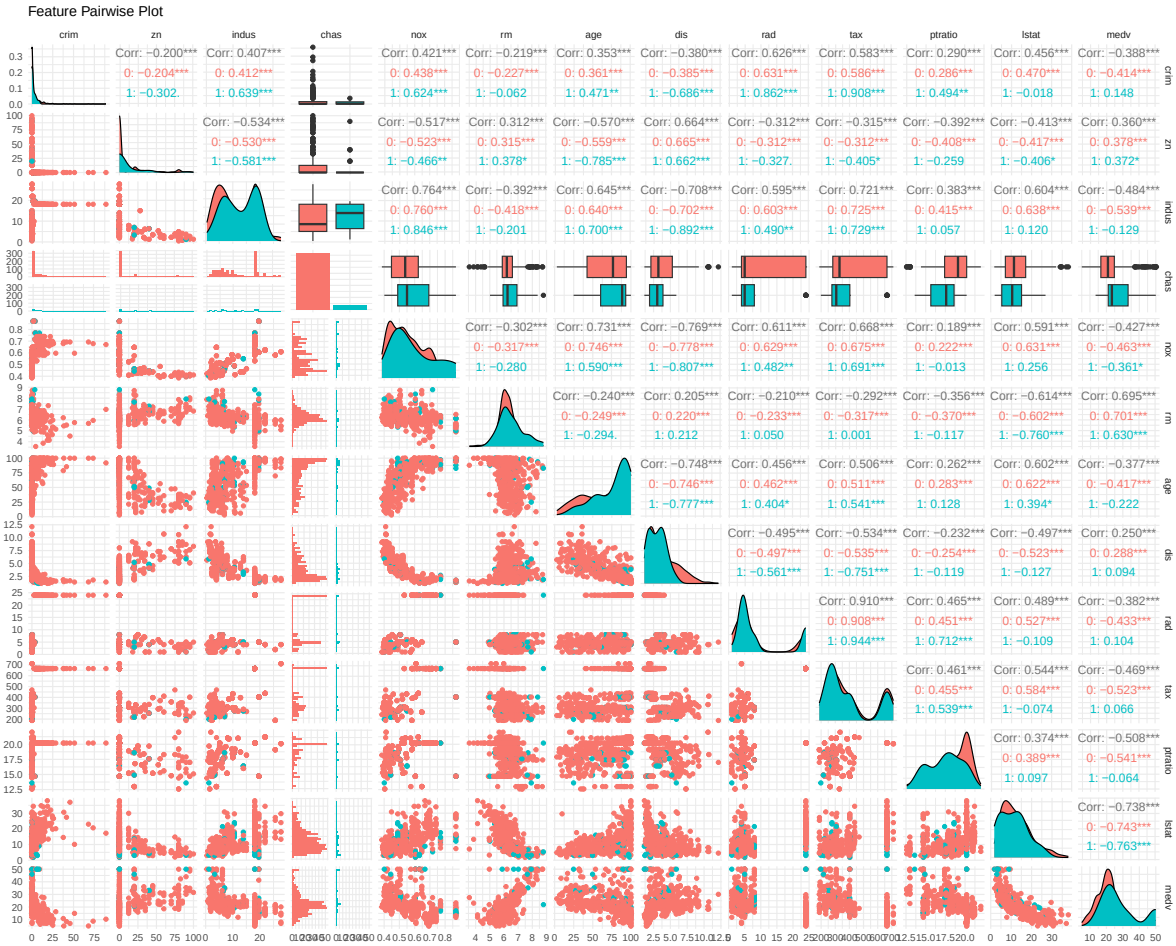


Figure 4: Feature Pairplot

Analyzing the significance of the correlations, the correlations between the response variable, medv, and all other numeric features appears to be significant at the 1% level. The observed relationship between medv and its highest correlated feature, lstat appears to be negatively non-linear.

Another interesting feature interaction is that of nox and the dis features which show a significant negative correlation, which the shape of the scatter plot also indication non-linearity. This intuitively reasonable as the greater the distance to employment centres- would indicate lower levels of pollutants in the the area, with levels also rapidly decreasing as the distances increase * word this better

Model Building

Baseline Model:All Feature Regression Summary

medv

crim -0.13***
(0.04)
zn 0.06***
(0.02)
indus 0.03
(0.08)
chas1 1.77*
(0.96)
nox -20.60***
(4.41)
rm 3.75***
(0.49)
age -0.01
(0.02)
dis -1.65***
(0.23)
rad 0.29***
(0.08)
tax -0.01***
(0.005)
ptratio -1.01***
(0.15)
lstat -0.48***
(0.06)
Constant 44.45***
(5.65)
N 400
R2 0.74
Adjusted R2 0.73
Residual Std. Error 4.84 (df = 387)
F Statistic 90.99*** (df = 12; 387)
==========
Notes: ***Significant at the 1 percent level.*** *Significant at the 5 percent level.* Significant at the 10 percent level.

Model Performance:

- The model shows a relatively good fit to the training data, with ~74% of the variation in the data, explained by the model
- 10 out of the 12 feature coefficients (excluding the intercept) are statistically different from 0 at at lest the 10% level, with 9 of these significant at the 1% level
- Nitrogen Oxide (nox) levelsshow the biggest marginal change , in absolute value, to median house values, medv, with an estimated negative change of -20.6K in estimated average houspe values per one unit increase in nox levels.
- The F statistic is also significant at the 1% level, indicating that the coefficients of the model are jointly statistically different from 0.

Prediction of test data set

Table 2: Baseline Test MSE

baseline.mse
16.64

The out-of-sample (OOO) MSE for the baseline model is at 16.64K, that is, on average the baseline model's home values predictions are expected to deviate from the actual by this value, which is greater that the standard deviation of median house values of ~9.2.

Model Improvement

Backwards Step wise Variable Selection

Method:

1. Start off with the baseline model: regress response variable against all features
2. Iteratively remove one feature, retrain the model and store both the AIC and BIC scores for each retrained model.

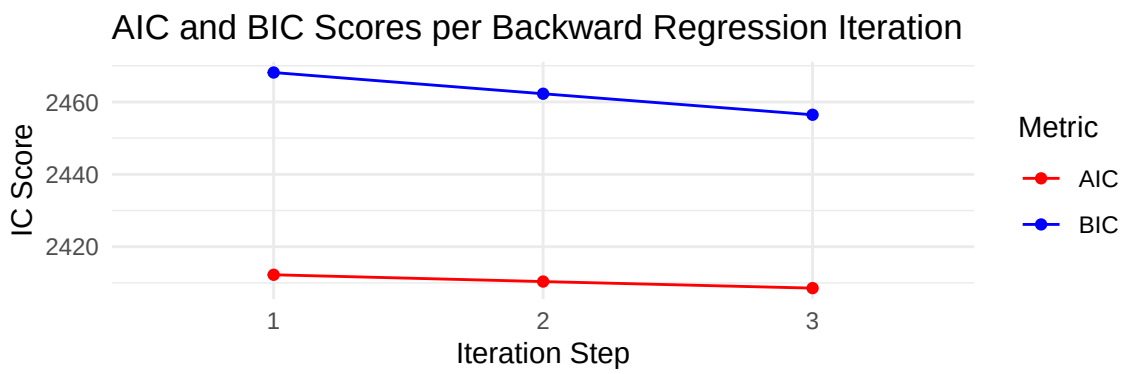


Figure 5: AIC and BIC curves per Backwise Steo Regression Iteration

```
Backward Stepwise Best Model Regression Summary
=========================================================
                                     medv
---------------------------------------------------------
crim                                -0.13***
                                     (0.04)
zn                                   0.06***
                                     (0.02)
chas1                                1.79*
                                     (0.95)
nox                                 -20.42***
                                     (4.05)
rm                                   3.70***
                                     (0.48)
dis                                 -1.65***
                                     (0.22)
rad                                  0.28***
                                     (0.07)
tax                                 -0.01***
                                    (0.004)
ptratio                             -1.01***
                                     (0.15)
lstat                               -0.49***
                                     (0.05)
Constant                            44.31***
                                     (5.61)
N                                     400
R2                                    0.74
```

```
Adjusted R2                          0.73
Residual Std. Error            4.83 (df = 389)
F Statistic                 109.63*** (df = 10; 389)
==========================================================
Notes:               ***Significant at the 1 percent level.
                      **Significant at the 5 percent level.
                       *Significant at the 10 percent level.
```

Backward Step Regression Results:

1. The step model reached the lowest AIC (~2408) and BIC(~2456) after 3 iterations with 10 out of the 12 features retained.
2. Retained Features: **crime, zn, chas, nox, rm, dis, rad, tax, ptratio and lstat** with the **age** and **indus** variables removed**.** All of these are still statistically significant at least up to the 10% level.
3. All regressors also maintain their sign with very little changes to their maginal effects with respect to the response variable.
4. R square (~74%) and the F statistic (significant at 1%) are still comparable and in the same region as the baseline model

Ridge and Lasso Regression

Method: Use cross validation to find the optimal regularization parameter(s) ,lambda, that produce the minimum CV MSE error during training and compare the coefficients produced by the model.

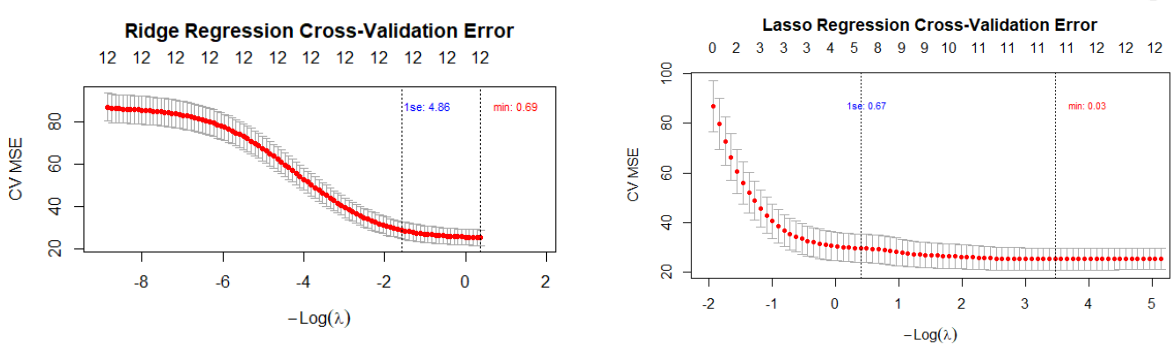


Figure 6: Cross-validation mean squared error plots for Ridge and Lasso regression.

Table 3: Ridge and LASSO Regression Coefficients

Feature	Ridge λ_{min}	Ridge λ_{1se}	LASSO λ_{min}	LASSO λ_{1se}
(Intercept)	34.7669663	25.4992007	42.5668194	20.4436614
crim	-0.0997802	-0.0704068	-0.1185652	-0.0232736
zn	0.0403742	0.0248068	0.0520495	0.0000000
indus	-0.0420514	-0.0746080	0.0000000	0.0000000
chas	2.0742555	2.1120254	1.7878073	0.7975187
nox	-13.8315658	-5.8133328	-19.3000339	-3.8251303
rm	3.9794124	3.4638733	3.7861161	4.3258353
age	-0.0082776	-0.0113347	-0.0039237	0.0000000
dis	-1.2318078	-0.4766797	-1.5880810	-0.1559487
rad	0.1208266	-0.0006445	0.2381425	0.0000000
tax	-0.0071776	-0.0037944	-0.0122039	-0.0020665
ptratio	-0.9140152	-0.6941726	-0.9950010	-0.8503920
lstat	-0.4340762	-0.3124781	-0.4792334	-0.4635408

Regression Results:

1. The best regularization parametrs for the Ridge and LASSO models are reached at at λ =0.69 (CV_MSE$_{\text{min}}$ of ~25.48) and 0.69 (CV_MSE$_{\text{min}}$ ~27.9), respectively . The 1_{se} equivalent parameters for Ridge and LASSO are obtained at λ=4.86 (CV_MSE$_{\text{min}}$ ~28.79) and λ=0.67 (CV_MSE$_{\text{min}}$ ~27.99).
2. For the simplest model, LASSO$_{\text{1SE}}$, 7 non-zero coefficients remained in the final model; the coefficients of **zn, indus, age,** and **rad** were all pushed to zero. In particular, **zn and rad** were statistically significant in the baseline model.
3. The **nox** feature still maintains its position as the largest absolute marginal predictor of median house values across three out of the four models, with the rm feature having the largest marginal contribution to response variable.
4. In addition to a change in magnitude, some feature coefficients also show a change in sign in some of models: the **indus** coefficient also sees a change (positive to negative) in sign from the baseline in both the Ridge$_{\text{1SE}}$ and Ridge$_{\text{min}}$ models while the **rad** feature coefficient changes sign in the Ridge$_{\text{1SE}}$ model (although this feature coefficient is highly penalized relative to the baseline).

Predictions On Test Data

For both the Ridge and LASSO regressions, the 1SE models will be used to favour model simplicity and reduce the risk of over fitting; these are used to predict on the test data.

Table 4: Test Data MSE Model Metrics

metric	all_features	backwise_step	ridge	lasso	average.across.models
MSE	16.6422	16.64976	16.96179	19.77408	14.00557

Although the test scores MSE are relatively close to each other for 3 out of the 4 models, the variable selection models fail to outperform the baseline model LASSO having the worst test mse out of the 4. * Talk about implications of multicollinearity in conclusion

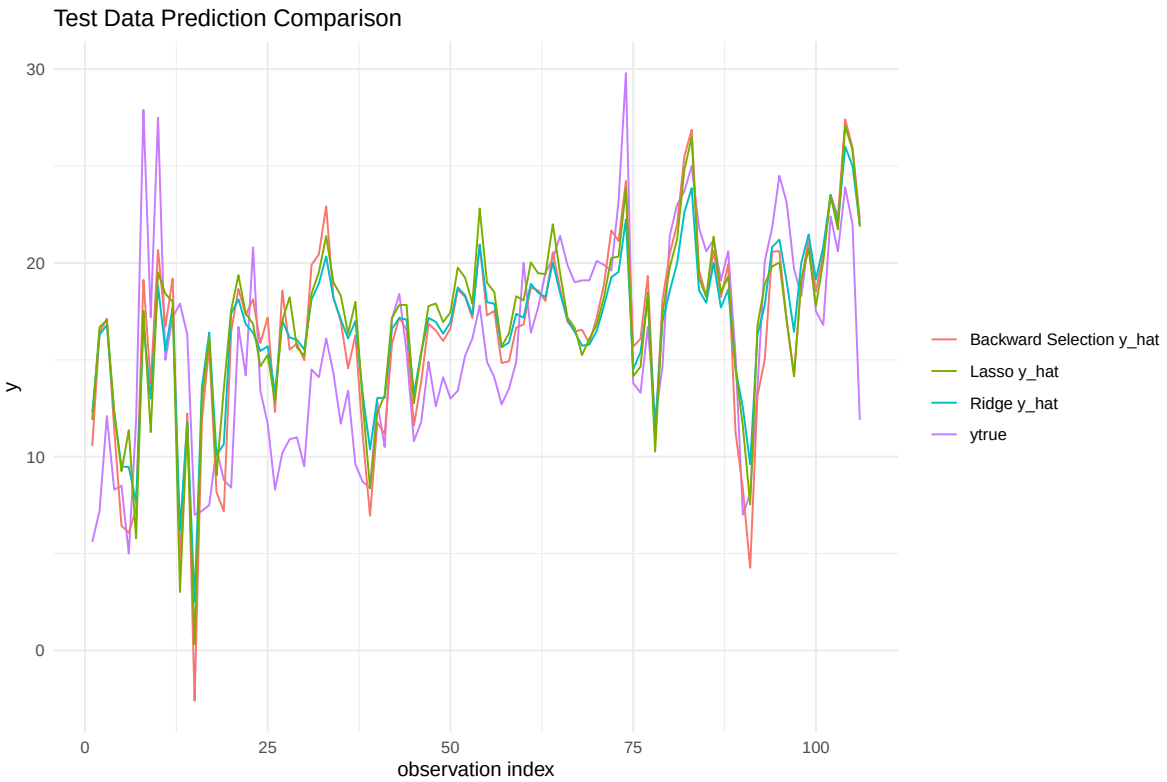
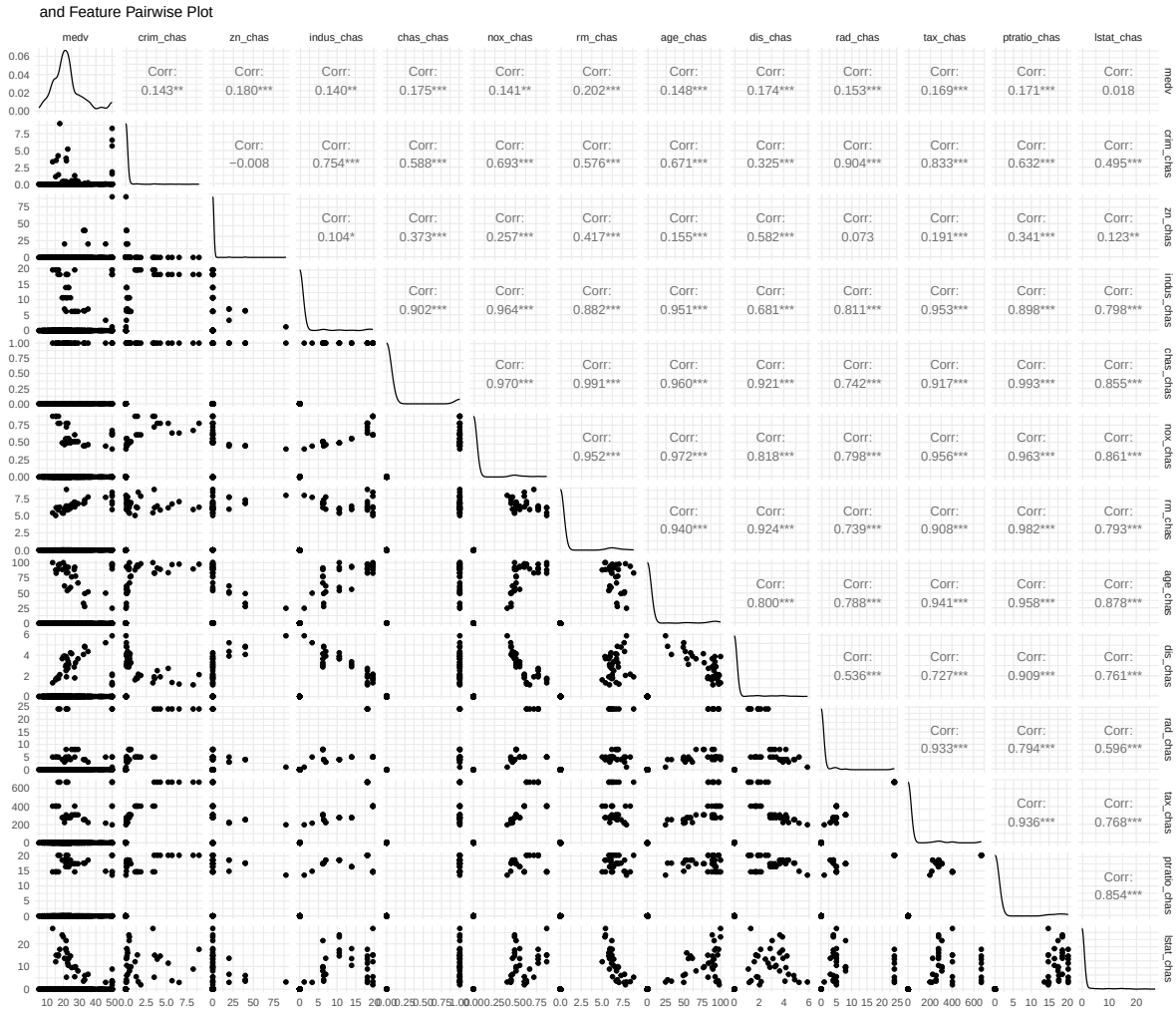


Figure 7: Predictions Vs Actual Values

All the model predictions show relatively good fit to the general shape of the response vatiale in the test data.

Inclusion of Interaction Term



Median House Values, medv, shows the highest correlation with the rm_chas interaction variable. Look at and this correlation also happens to be statistically significant at 5%.

```
Call:
lm(formula = medv ~ ., data = int_tr_d)

Residuals:
    Min      1Q  Median      3Q     Max
-13.175  -2.938  -0.783   1.849  27.677

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  44.526464   5.733305   7.766 7.36e-14 ***
```

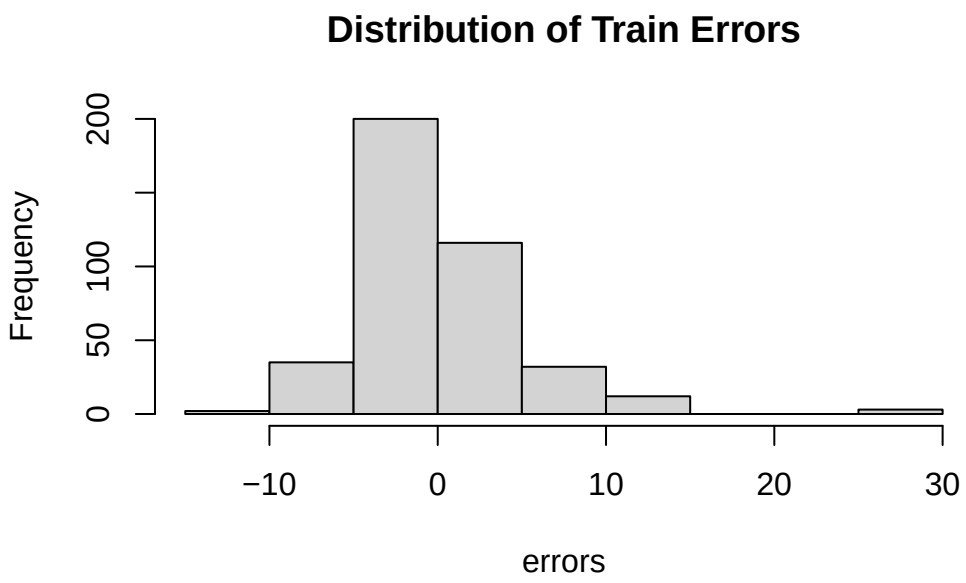
```
crim         -0.127376   0.039281  -3.243 0.001287 **
zn            0.055996   0.015879   3.526 0.000472 ***
indus         0.032937   0.077219   0.427 0.669953
chas          1.184828   7.117119   0.166 0.867870
nox         -20.577159   4.420878  -4.655 4.47e-06 ***
rm            3.734252   0.510945   7.309 1.57e-12 ***
age          -0.005220   0.015165  -0.344 0.730850
dis          -1.650745   0.232964  -7.086 6.61e-12 ***
rad           0.286979   0.079776   3.597 0.000363 ***
tax          -0.014810   0.004696  -3.154 0.001739 **
ptratio      -1.014128   0.147635  -6.869 2.60e-11 ***
lstat        -0.480899   0.057936  -8.301 1.76e-15 ***
rm_chas       0.088848   1.076285   0.083 0.934251
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.851 on 386 degrees of freedom
Multiple R-squared:  0.7383,    Adjusted R-squared:  0.7295
F-statistic: 83.77 on 13 and 386 DF,  p-value: < 2.2e-16
```

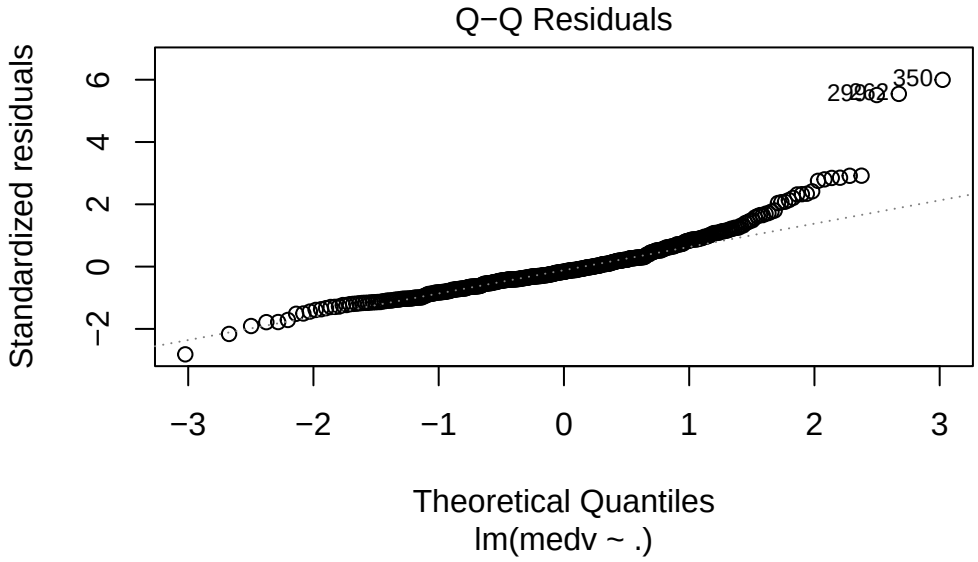
rm_chas did not turn out to be significant- but I'm not sure if this is as a result of the data imbalance (lot more chas=0 than 1) ?

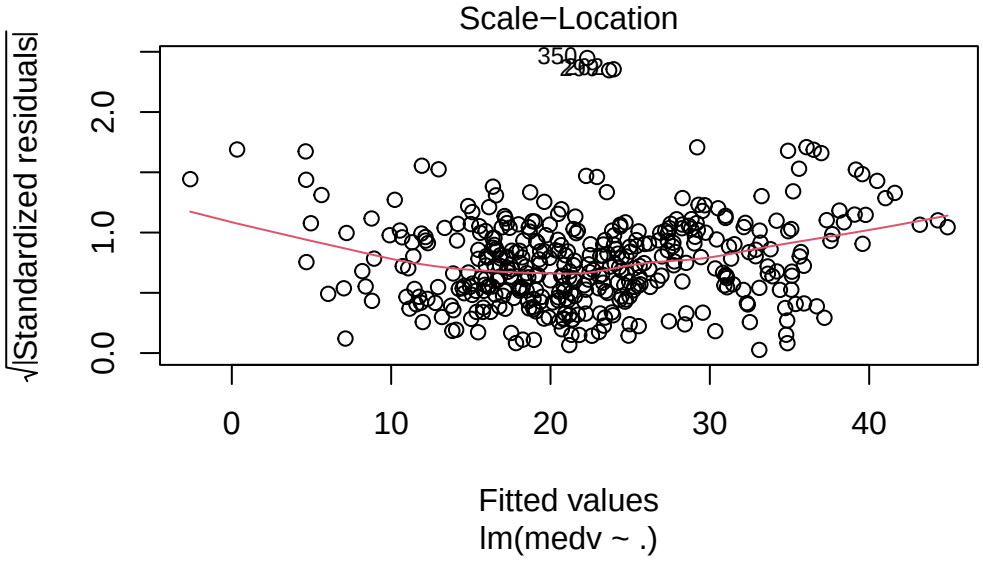
```
[1] 16.61995
```

```
[1] 16.6422
```

Residual Diagnostics




Q−Q Residuals
Standardized residuals
Theoretical Quantiles
lm(medv ~ .)
350


Scale−Location
√|Standardized residuals|
Fitted values
lm(medv ~ .)
350

Appendix

ref website:https://alain-vandormael.netlify.app/post/varselect/#:~:text=Stepwise%20regression,next%20deletion%20increases%20the%20AIC.

https://statisticsbyjim.com/regression/interaction-effects/